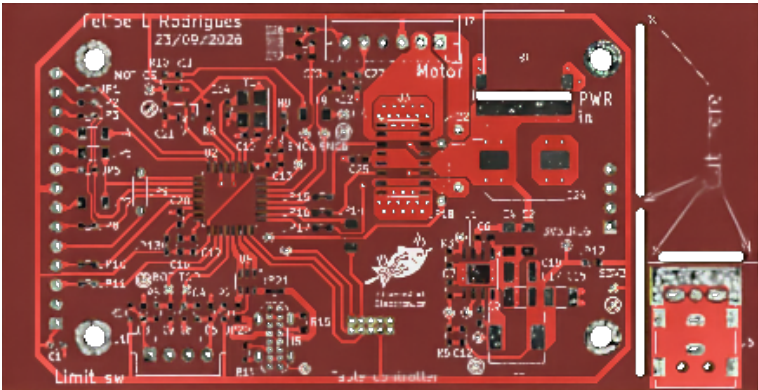
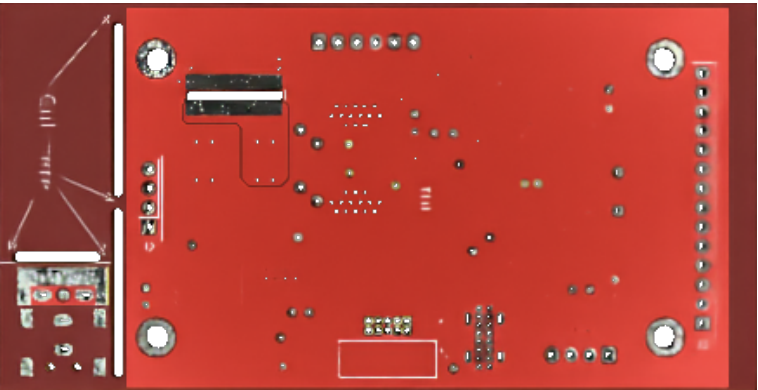


Desk controller fabrication drawing

Top view



Bottom view



	Desk controller		Sheet: Bare board overview File: ktest.kicad_pcb	
	Notes		Rev: 0.0.2	Drwn: FLR
	c1		Mat: Concept	Rev'd: FLR
	c2		Date: 06/10/2026	RI's'd: FLR
	c3		Dwg: NA	Size: A4
	c4		Generated with KiCad	Page: 1/14

Fabrication requirements

Fabrication requirements

IPC

In case of divergences, dimensions from Gerber files take precedence

Surface finish: Lead-free HASL

Copper thickness: minimum x um external layer, minimum x internal layer, per IPC xxx

vias to be left untented / tented / filled and capped

process must be RoHS compliant

flammability rating shall meet as a minimum ul94v-0

tg170 or better

tolerance xxx or as per ipc

Board build up

	Material	Layer	Thickness	Dielectric	Type
		F.Paste			Paste Mask
		F.Silkscreen			Legend
		F.Mask	0.01mm		Solder Mask
	Copper	F.Cu	0.035mm (1.00oz)		Signal
	Prepreg		0.1mm	FR4	Dielectric
	Copper	In1.Cu	0.035mm (1.00oz)		Signal
	Core		1.24mm	FR4	Dielectric
	Copper	In2.Cu	0.035mm (1.00oz)		Signal
	Prepreg		0.1mm	FR4	Dielectric
	Copper	B.Cu	0.035mm (1.00oz)		Signal
		B.Mask	0.01mm		Solder Mask
		B.Silkscreen			Legend
		B.Paste			Paste Mask

Total thickness: 1.6mm



Desk controller

Notes

c1
c2
c3
c4

Sheet: Fabrication requirements

File: ktest.kicad_pcb

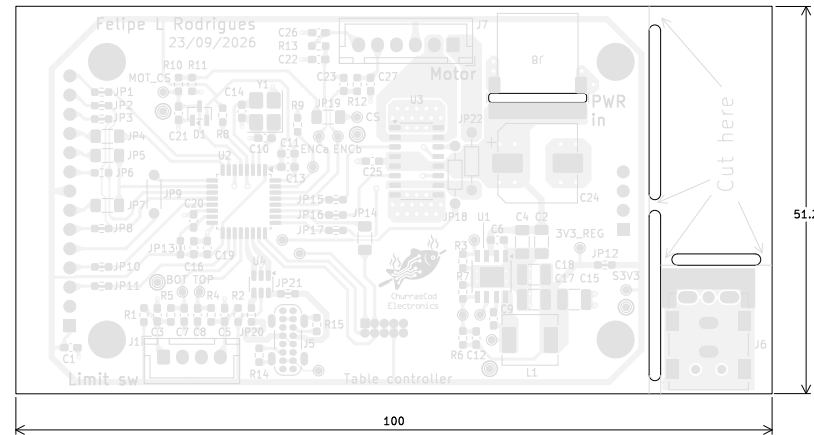
Rev: 0.0.2
Mat: Concept
Date: 06/10/2026

Dwg: NA
Generated with KiCad

Drwn: FLR
Rev'd: FLR
Rls'd: FLR

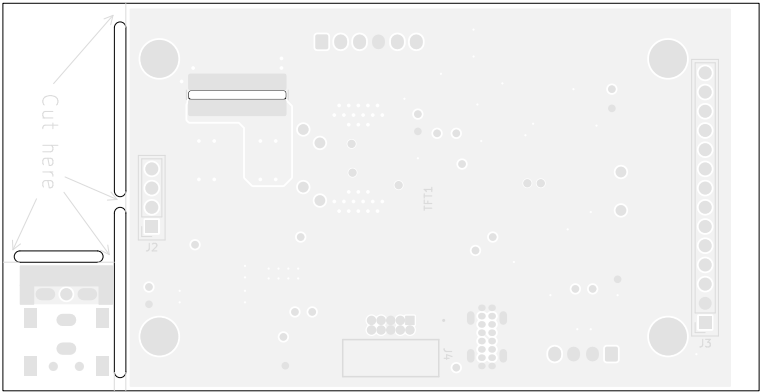
Size: A4
Page: 2/14

Top fabrication (scale 1:1)



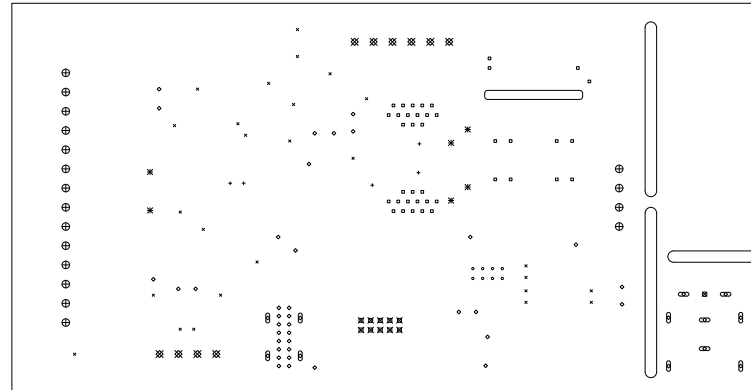
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	c2		Mat: Concept	Rev'd: FLR
	c3		Date: 06/10/2026	RLs'd: FLR
c4	Generated with KiCad		Dwg: NA	Size: A4
				Page: 3/14

Bottom fabrication (scale 1:1)



Desk controller		Sheet: Bottom fabrication (scale 1:1)	
		File: ktest.kicad_pcb	
Notes			Drwn: FLR
			Rev'd: FLR
			RLs'd: FLR
			Size: A4
		Generated with KiCad	Page: 4/14

Drill drawing — L1 - L4 (scale 1:1)



Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair
×	26	0.30mm	PTH	Round	F.Cu – B.Cu
○	8	0.30mm	PTH	Round	F.Cu – B.Cu
+	5	0.40mm	PTH	Round	F.Cu – B.Cu
□	40	0.50mm	PTH	Round	F.Cu – B.Cu
◇	37 + 12	0.50mm	PTH	Round + Slot	F.Cu – B.Cu
⊗	1	0.60mm	PTH	Round	F.Cu – B.Cu
✱	6	0.70mm	PTH	Round	F.Cu – B.Cu
⊗	10	0.74mm	PTH	Round	F.Cu – B.Cu
⊗	10	0.95mm	PTH	Round	F.Cu – B.Cu
⊕	18	1.00mm	PTH	Round	F.Cu – B.Cu
Total 173					



Desk controller

Notes

c1
c2
c3
c4

Sheet: Drill drawing — L1 - L4 (scale 1:1)

File: ktest.kicad_pcb

Rev: 0.0.2
Mat: Concept
Date: 06/10/2026

Dwg: NA
Generated with KiCad

Drwn: FLR
Rev'd: FLR
Rls'd: FLR
Size: A4
Page: 5/14

Drill drawing — L1 - L4 (scale 1:1)



Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair
×	2	1.20mm	NPTH	Round	F.Cu – B.Cu
○	4	3.20mm	NPTH	Round	F.Cu – B.Cu
	Total 6				



Desk controller

Notes

c1
c2
c3
c4

Sheet: Drill drawing — L1 - L4 (scale 1:1)

File: ktest.kicad_pcb

Rev: 0.0.2

Mat: Concept

Date: 06/10/2026

Dwg: NA

Generated with KiCad

Drwn: FLR

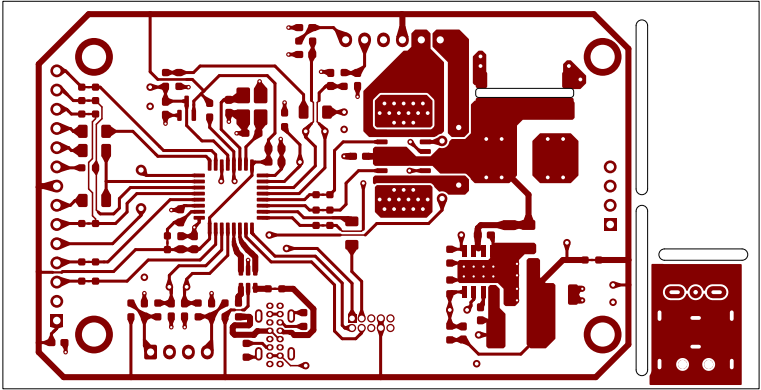
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RLs'd: FLR

Size: A4

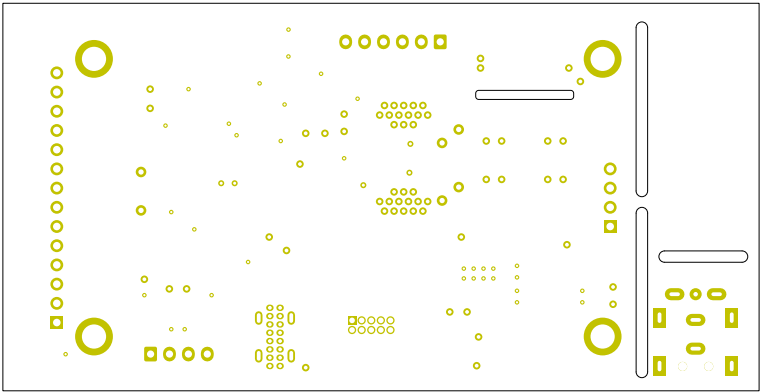
Page: 6/14

F.Cu (scale 1:1)



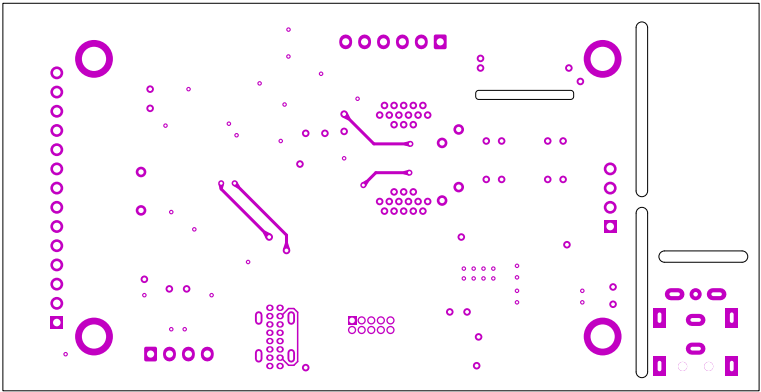
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Notes c1 c2 c3 c4	Rev: 0.0.2		Drwn: FLR
	Mat: Concept		Rev'd: FLR
	Date: 06/10/2026		RLs'd: FLR
	Dwg: NA		Size: A4
		Generated with KiCad	Page: 7/14

In1.Cu (scale 1:1)



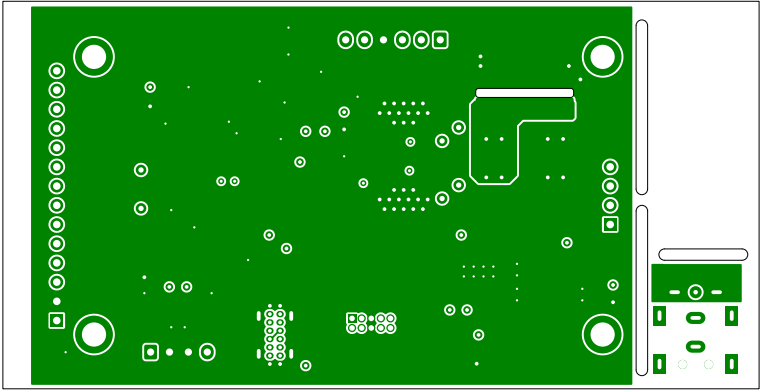
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Notes c1 c2 c3 c4	Rev: 0.0.2		Drwn: FLR
	Mat: Concept		Rev'd: FLR
	Date: 06/10/2026		RLs'd: FLR
	Dwg: NA		Size: A4
		Generated with KiCad	Page: 8/14

In2.Cu (scale 1:1)



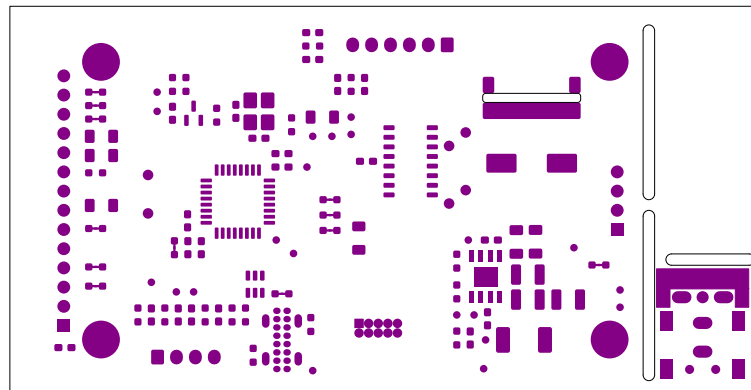
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Notes c1 c2 c3 c4	Rev: 0.0.2		Drwn: FLR
	Mat: Concept		Rev'd: FLR
	Date: 06/10/2026		RLs'd: FLR
	Dwg: NA		Size: A4
		Generated with KiCad	Page: 9/14

B.Cu (scale 1:1)



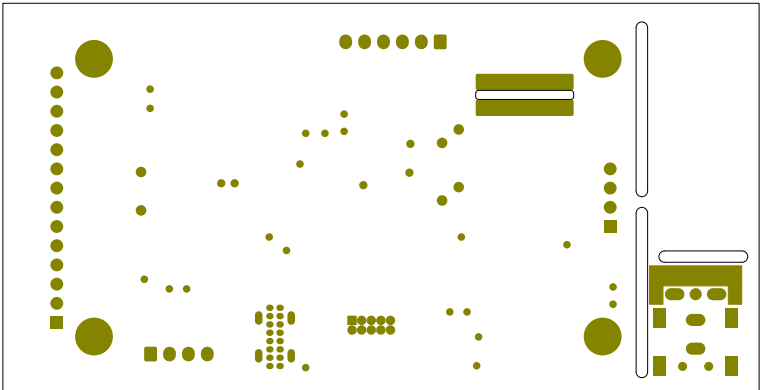
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Notes c1 c2 c3 c4		File: ktest.kicad_pcb	Drwn: FLR
		Rev: 0.0.2	Rev'd: FLR
		Mat: Concept	RLs'd: FLR
		Date: 06/10/2026	Size: A4
		Dwg: NA	Page: 10/14
		Generated with KiCad	

F.Mask (scale 1:1)



	Desk controller		Sheet: F.Mask (scale 1:1)	
	Notes		File: ktest.kicad_pcb	
	c1		Rev: 0.0.2	Drwn: FLR
	c2		Mat: Concept	Rev'd: FLR
	c3		Date: 06/10/2026	RLs'd: FLR
c4	Dwg: NA		Size: A4	
	Generated with KiCad		Page: 11/14	

B.Mask (scale 1:1)



Desk controller		Sheet: B.Mask (scale 1:1)	
		File: ktest.kicad_pcb	
Notes c1 c2 c3 c4		Rev: 0.0.2	Drwn: FLR
		Mat: Concept	Rev'd: FLR
		Date: 06/10/2026	RLs'd: FLR
		Dwg: NA	Size: A4
		Generated with KiCad	Page: 12/14

Felipe L Rodrigues
23/09/2026

C26
R13
C22

J7

8r

Motor

PWR in

Cut here

JP12

S3V3

J6

Limit sw

Table controller

Churned Electronics

JP18 U1

C4 C2

3V3_REG

JP12

C18 C15

S3V3

C9

R6 C12

L1

J5

R15

JP20

C3 C7 C8 C5

R14

J1

R1

JP11

B07 TOP

R5

R4

R2

JP13

C16

C19

JP10

JP8

JP7

JP9

C20

JP6

JP5

JP4

C21

D1

R8

U2

JP3

JP2

JP1

MOT_CS

R10

R11

Y1

C14

C23

C27

ENCa

ENCb

C11

C10

C13

C25

JP15

JP16

JP17

JP14

U3

CS

JP19

R12

C5

U4

JP21

R9

JP19

R12

C5

U3

CS

JP22

C24

C27

C23

C26

R13

C22

J7

8r

PWR in

Cut here

JP12

S3V3

J6

Limit sw

Table controller

Churned Electronics

JP18 U1

C4 C2

3V3_REG

JP12

C18 C15

S3V3

C9

R6 C12

L1

J5

R15

JP20

C3 C7 C8 C5

R14

J1

R1

JP11

B07 TOP

R5

R4

R2

JP13

C16

C19

JP10

JP8

JP7

JP9

C20

JP6

JP5

JP4

C21

D1

R8

U2

JP3

JP2

JP1

MOT_CS

R10

R11

Y1

C14

C23

C27

ENCa

ENCb

C11

C10

C13

C25

JP15

JP16

JP17

JP14

U3

CS

JP19

R12

C5

U4

JP21

R9

JP19

R12

C5

U3

CS

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C24

C27

C23

C26

R13

C22

J7

8r

PWR in

Cut here

JP12

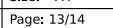
S3V3

J6

Limit sw

Table controller

Churned Electronics



B.Silkscreen (scale 1:1)

12

13

14

15

16

Cut hole

	Desk controller		Sheet: B.Silkscreen (scale 1:1)	
			File: ktest.kicad_pcb	
	Notes		Rev: 0.0.2	Drwn: FLR
	c1		Mat: Concept	Rev'd: FLR
	c2		Date: 06/10/2026	RLs'd: FLR
c3		Dwg: NA	Size: A4	
c4		Generated with KiCad	Page: 14/14	



Notes

c1
c2
c3
c4

File: ktest.kicad_pcb

Rev:	0.0.2
Mat:	Concept
Date:	06/10/2020

Dwg: NA
Generated with KiCad

Drwn: FLR
Rev'd: FLR
RI's'd: FLR
Size: A4
Page: 14/14